

ESP32 Adaptive Smart Lighting System

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Projektbeschreibung

Dieses Projekt demonstriert ein intelligentes adaptives Beleuchtungssystem mit ESP32, PWM-Technologie und einem LDR-Lichtsensord. Der ESP32 misst kontinuierlich die Lichtintensität der Umgebung und steuert automatisch die Helligkeit der LEDs. Bei Dunkelheit werden die LEDs heller, während sie bei starkem Licht schwächer werden oder sich vollständig ausschalten. Die grüne und rote LED sind parallel mit GPIO14 verbunden und reagieren gleichzeitig auf Veränderungen der Umgebungsbeleuchtung.

Verwendete Komponenten

Komponente	Beschreibung
ESP32 DevKit V1	WiFi Mikrocontroller
LDR Sensor	Lichtintensität messen
Grüne LED	PWM adaptive Beleuchtung
Rote LED	Parallel verbundene LED
Widerstände	Strombegrenzung
Breadboard	Elektronische Verbindungen
MicroPython	Programmiersprache

Systemverhalten

- Dunkelheit → LEDs werden heller.
- Mittlere Beleuchtung → LEDs leuchten schwächer.
- Helle Umgebung → LEDs schalten sich aus.
- PWM ermöglicht sanfte Helligkeitsübergänge.

MicroPython Code

```
from machine import Pin, ADC, PWM
import time

green_led = PWM(Pin(14))
green_led.freq(1000)

internal_led = Pin(2, Pin.OUT)
```

```
light_sensor = ADC(Pin(34))
light_sensor atten(ADC.ATTN_11DB)
light_sensor.width(ADC.WIDTH_12BIT)

threshold = 1800

while True:
    light_value = light_sensor.read()

    if light_value < 1800:
        brightness = 0
        internal_led.off()

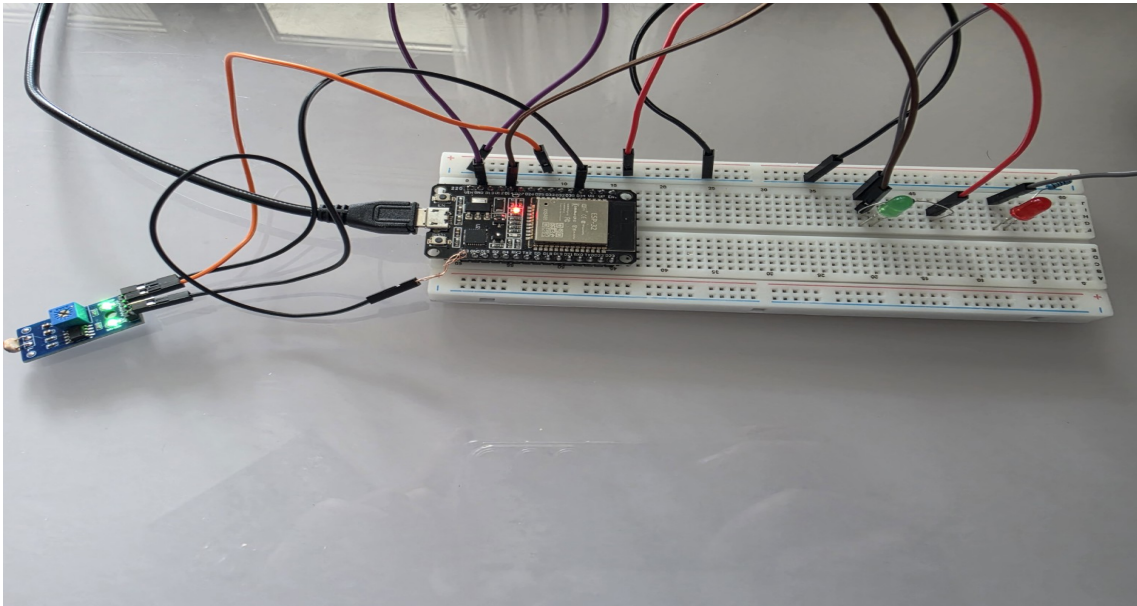
    elif light_value < 2300:
        brightness = 400
        internal_led.on()

    else:
        brightness = 1023
        internal_led.on()

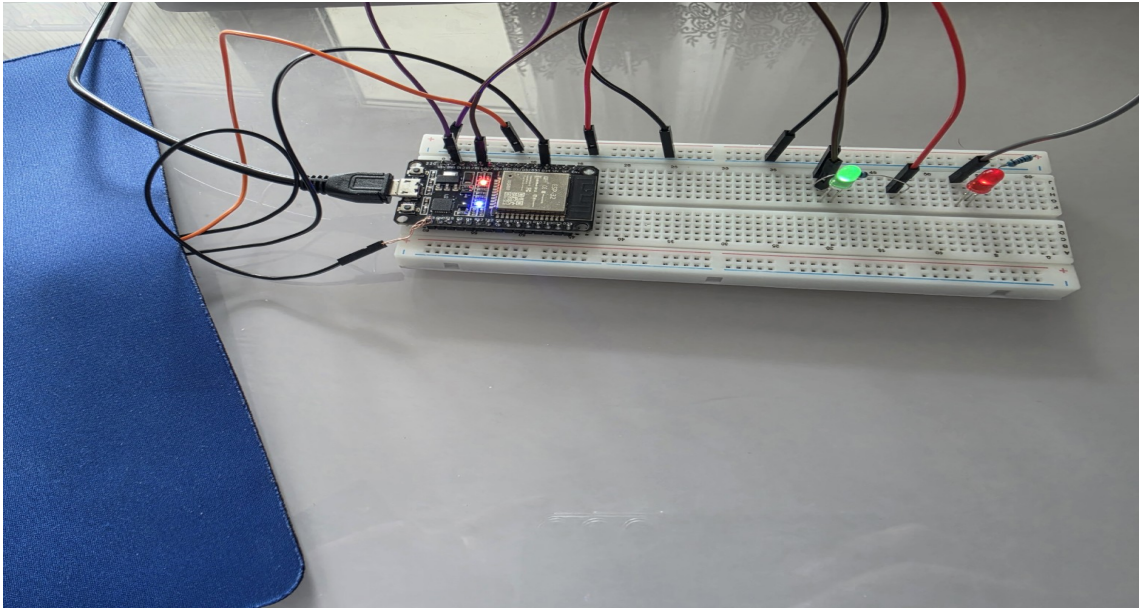
    green_led.duty(brightness)
    time.sleep(0.5)
```

Projektbilder

Helle Umgebung → LEDs ausgeschaltet



Dunkelheit → LEDs eingeschaltet



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Project Description

This project demonstrates an intelligent adaptive lighting system using ESP32, PWM technology, and an LDR light sensor. The ESP32 continuously monitors ambient light intensity and automatically controls LED brightness. In dark environments, the LEDs become brighter, while in bright environments they become dimmer or turn OFF completely. The green LED and red LED are connected in parallel to GPIO14 and react simultaneously to environmental light changes.

Components Used

Component	Description
ESP32 DevKit V1	WiFi-enabled microcontroller
LDR Sensor	Measures light intensity
Green LED	PWM adaptive lighting
Red LED	Parallel connected LED
Resistors	Current limiting
Breadboard	Electronic connections
MicroPython	Programming language

System Behavior

- Dark environment → LEDs become brighter.
- Medium light → LEDs operate with dim brightness.
- Bright environment → LEDs turn OFF.
- PWM provides smooth brightness transitions.

MicroPython Code

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import time

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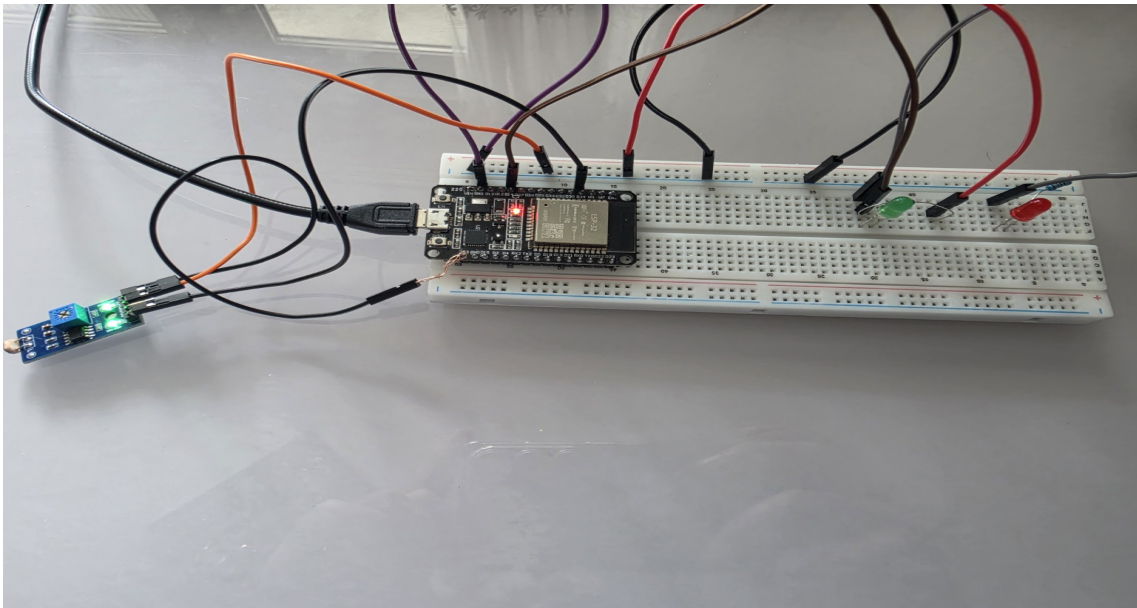
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    else:
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    green_led.duty(brightness)
    time.sleep(0.5)
```

Project Images

Bright Environment → LEDs OFF



Dark Environment → LEDs ON

